

Algebra II with Trigonometry

Chapter 7.2

Olympiad Problems

Saved tutoring response with MathJax rendering. Original PDF file:
[4582337_alg-2trig-h-chp-7-2-note-docx.pdf](#)

Problems

1. Find the exact value of $\cos 75^\circ + \sin 15^\circ$.
2. Without a calculator, evaluate $\tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{1}{3}\right)$.
3. Simplify $\frac{\sin(x+y) + \sin(x-y)}{\cos(x+y) + \cos(x-y)}$ to an expression in a single trigonometric function.
4. Express $7 \sin \theta - 24 \cos \theta$ in the form $R \sin(\theta - \phi)$, then find θ in $0^\circ < \theta < 360^\circ$ such that $7 \sin \theta - 24 \cos \theta = 25$.
5. If $\sin \alpha = \frac{5}{13}$ with α in the first quadrant, and $\cos \beta = -\frac{3}{5}$ with β in the second quadrant, mentally compute $\sin(\alpha + \beta)$.
6. Prove that for all real x , $\cos(x) - \cos\left(x + \frac{2\pi}{3}\right) + \cos\left(x + \frac{4\pi}{3}\right) = 0$.